ASISH OLIVER M

MOTORSPORT SOFTWARE ENGINEER | AI / ML ENGINEER

Trichy, Tamil Nadu, India * dreamteamoliver@gmail.com * github.com/T-Bone47 * linkedin.com/in/asish-oliver-56b454324

ENGINEERING PROFILE

Motorsport Software Engineer and AI/ML Systems Engineer specializing in high-throughput real-time telemetry pipelines, numerical quasi-steady vehicle dynamics simulation, race strategy modeling, and applied machine learning. Committed to zero-fabrication engineering rigor, mathematical verification, strict architectural decoupling, and clean systems design.

EDUCATION

VIT-AP UNIVERSITY **2021 - Present**

Integrated M.Tech in Computer Science and Engineering

Core Coursework: Distributed Systems, High-Performance Computing, Machine Learning, Numerical Methods, Software Architecture.

ENGINEERING EXPERIENCE

Independent Motorsport / eSports **2025 - Present**

Race & Strategy Engineer

* Architected game-agnostic UDP telemetry acquisition pipeline ingesting high-frequency F1 vehicle motion and physics packets.
* Formulated dynamic stint degradation models & regression curves estimating thermal wear cliff and optimal pit-stop windows.
* Built AI Race Engineer reasoning engine providing automated fuel, tire delta, and safety car tactical advisories.

IGNICT **2026 - Present**

AI / Machine Learning Intern

* Engineered end-to-end Python data preprocessing pipelines, feature engineering, and model evaluation routines.
* Researched unsupervised player representation vectors and dimensionality reduction architectures using PCA and t-SNE.
* Developed offline-first speech-to-text inference integration (Vosk) and time-series demand forecasting models (XGBoost).

CANONICAL ENGINEERING SYSTEMS

Live F1 Intelligence | High-Throughput Distributed Telemetry Pipeline Python, FastAPI, TimescaleDB, WebSockets, Docker

* Ingests 4 live feeds (OpenF1, FastF1, Jolpica, SignalR), validates schemas, and normalises events into a single canonical model.
* Benchmark Evidence: 1,067,193 canonical events processed, 0 malformed records, ~2,650 events/s, ~0.10 ms p50, 99 MB peak memory.

F1 Lap-Time Simulator & Lap Prediction | Numerical Vehicle Dynamics Python, NumPy, SciPy, Matplotlib

* Formulated quasi-steady-state numerical lap solver utilizing G-G friction circles, aero downforce mapping, and grip envelopes.
* Computes forward acceleration and backward braking numerical integration passes to solve theoretical minimum lap times.

RaceMind-AI | Real-Time Race Strategy & UDP Telemetry Bridge Python, UDP Sockets, Streamlit, Scikit-learn

* Game-agnostic telemetry abstraction layer decoding multi-generation F1 UDP packets (velocities, sector deltas, tire temperatures).
* Implemented rolling stint wear regression and live AI Race Engineer contextual query terminal for undercut/overcut decisions.

F1 Race Manager | F1 Team Management Simulation Game Engine TypeScript, React, Next.js, State Machine Architecture

* Deterministic race simulation architecture featuring lap-by-lap pace calculation, weather radar volatility, and safety car triggers.

Football Intelligence OS | 128D Player Representation Vector Engine Python, Pandas, Scikit-learn, PCA, Cosine Metric

* Cleaned and normalized 16,228 FC26 players into a dense 128D latent embedding space; serves tactical similarity replacements.

VyaparPulse | Offline-First Voice Ledger & XGBoost Demand Forecasting Python, FastAPI, Vosk STT, XGBoost, SQLite

* Offline voice-to-structured ledger committing transactions in <800 ms; 7-day revenue forecasting with cold-start fallbacks.

TECHNICAL ARSENAL

Languages: Python 3.12, TypeScript, MATLAB, C#, SQL, HTML5 / CSS3
Systems & Backend: FastAPI, Next.js 15, React 19, WebSockets, UDP Sockets, Docker, Microservices Architecture
Data & Time-Series: TimescaleDB, PostgreSQL, SQLite, SQLAlchemy, Pandas, NumPy, FastF1, OpenF1 Ingestion
AI / Machine Learning: Scikit-learn, XGBoost, PyTorch, Vosk Offline STT, PCA, t-SNE Embeddings, Time-Series Forecasting
Motorsport Engineering: Vehicle Dynamics Modeling, Quasi-Steady Lap Solver, G-G Friction Circle, Tyre Degradation Regression
3D & Visualization: Three.js, React Three Fiber, WebGL Performance Profiling, Voxel Engine Architecture